PRN:

Batch:

Experiment no 1

Title: Introduction to HPC Environment and Shared Memory Parallel Programming

Objective: To understand the HPC environment and implement basic programs using OpenMP.

Problem Statement:

High Performance Computing (HPC) involves the use of parallel processing techniques to solve computationally intensive problems. Study basics of OpenMP for shared memory parallel programming using CPU threads.

Task 1: HPC Environment Setup and System Configuration

Perform the following activities:

1. Identify and record your system configuration:
    - Processor model
    - Number of CPU cores
    - Number of logical processors
    - RAM size
2. Verify the installation using Linux OS and working of:
    - GCC compiler
    - OpenMP
3. Execute basic verification programs for each environment eg. Hello world program using multiple threads.

Task 2: OpenMP Thread Creation

Write an OpenMP program to create multiple threads and display thread information.

Requirements:

The program should:

1. Create a parallel region using OpenMP for finding square of first 1000 whole numbers.
2. Display:
   - Thread ID
   - Total number of threads created
3. Execute the program using:
   - 2 threads
   - 4 threads
   - 8 threads
   - 16 threads (if available)

Calculate speedup using:

$$Speedup = \frac{Sequential\ Execution\ Time}{Parallel\ Execution\ Time}$$

Task 3: Calculate Theoretical Peak FLOPS of CPU

Theoretical CPU performance can be estimated using:

$$\text{Peak FLOPS} = \text{Number of Cores} \times \text{Clock Frequency} \times \text{FLOP/Clock}$$

where:

- Number of cores = Physical CPU cores
- Clock frequency = CPU operating frequency in Hz
- FLOP/Clock = Number of floating-point operations performed per clock cycle

For SIMD-based CPUs:

$$\text{FLOPS} = \text{Cores} \times \text{Frequency} \times \text{SIMD Width} \times \text{Operations per Instruction}$$

Instruction Set	Precision	SIMD Width
SSE	Single Precision	4
AVX	Single Precision	8
AVX2	Single Precision	8
AVX-512	Single Precision	16

Where:

For Fused Multiply Add (FMA):

$$1\ \text{FMA} = 2\ \text{FLOPs}$$

(one multiplication + one addition)

Therefore:

$$\text{Peak FLOPS} = \text{Cores} \times \text{Frequency} \times \text{SIMD Width} \times 2$$

Compare the calculated flops with the manufacturers published Flops.